QUBO Formulation: Minimum Vertex Cover

Minimum Vertex Cover

Problem Statement. Given an undirected graph $G = (V, E)$ with vertex set V and edge set E, the Minimum Vertex Cover problem seeks a subset $S \subseteq V$ of minimum cardinality such that every edge in E is incident to at least one vertex in S. Formally, we wish to minimize $|S|$ subject to the constraint that for every edge $(u, v) \in E$, the intersection $\{u, v\} \cap S$ is non-empty.

Decision Variables. We introduce a binary variable $x_i \in \{0, 1\}$ for each vertex $i \in V$. The variable x_i indicates membership in the cover:

$$x_i = \begin{cases} 1 & \text{if } i \in S, \\ 0 & \text{if } i \notin S. \end{cases}$$

The size of the cover is given by $\sum_{i \in V} x_i$.

QUBO Derivation. We construct a Quadratic Unconstrained Binary Optimization (QUBO) Hamiltonian $H(x)$ whose global minimum over $\{0, 1\}^{|V|}$ corresponds to the optimal vertex cover.

Step 1: Objective Function. The objective is to minimize the number of selected vertices:

$$\min \sum_{i \in V} x_i.$$

In the QUBO formulation, this contributes directly to the Hamiltonian as a linear term with positive coefficients:

$$H_{\text{obj}}(x) = \sum_{i \in V} x_i. \tag{1}$$

Step 2: Constraints. For each edge $(u, v) \in E$, the vertex cover constraint requires that at least one endpoint is selected:

$$x_u + x_v \geq 1, \quad \forall (u, v) \in E.$$

This constraint is violated if and only if $x_u = 0$ and $x_v = 0$.

Step 3: Penalty Terms. To enforce the constraints, we add a penalty term for each edge that is zero when the constraint is satisfied and positive otherwise. A suitable penalty for the constraint $x_u + x_v \geq 1$ is:

$$P_{uv}(x) = \lambda(1 - x_u)(1 - x_v),$$

where $\lambda > 0$ is a penalty weight. Expanding this expression yields:

$$\begin{aligned} P_{uv}(x) &= \lambda(1 - x_u - x_v + x_u x_v) \\ &= \lambda - \lambda x_u - \lambda x_v + \lambda x_u x_v. \end{aligned} \tag{2}$$

Note that since $x_i \in \{0, 1\}$, we have $x_i^2 = x_i$, but no quadratic terms arise in the expansion of $(1-x_u)(1-x_v)$. The penalty is λ when $x_u = x_v = 0$ and 0 otherwise, correctly penalizing only infeasible assignments.

Step 4: Combined Hamiltonian. The total Hamiltonian is the sum of the objective and all edge penalties:

$$H(x) = \sum_{i \in V} x_i + \sum_{(u,v) \in E} \left(\lambda - \lambda x_u - \lambda x_v + \lambda x_u x_v\right). \tag{3}$$

We collect terms by grouping the constant offset, linear terms, and quadratic terms. The constant offset is:

$$c = \sum_{(u,v) \in E} \lambda = \lambda |E|. \tag{4}$$

The coefficient of a linear variable x_i is 1 from the objective plus contributions from each incident edge. Let $\deg(i)$ denote the degree of vertex i in G. Each incident edge (i, j) contributes $-\lambda x_i$. Thus:

$$\text{Coeff}(x_i) = 1 - \lambda \cdot \deg(i). \tag{5}$$

The coefficient of a quadratic term $x_u x_v$ is λ if $(u, v) \in E$, and 0 otherwise.

Step 5: QUBO Matrix Entries. The Hamiltonian can be written in standard QUBO form $H(x) = c + \sum_{i \in V} Q_{i,i} x_i + \sum_{(u,v) \in E} Q_{u,v} x_u x_v$. The entries of the upper-triangular Q matrix and the offset are given by:

$$Q_{i,i} = 1 - \lambda \cdot \deg(i), \quad \forall i \in V, \tag{6}$$

$$Q_{u,v} = \lambda, \quad \forall (u, v) \in E, \tag{7}$$

$$Q_{u,v} = 0, \quad \forall (u, v) \notin E, \tag{8}$$

$$c = \lambda |E|. \tag{9}$$

Penalty Weight Justification. The penalty weight λ must be chosen large enough to ensure that any infeasible solution has strictly higher energy than the best feasible solution. Consider a feasible solution x^{feas} and an infeasible solution x^{infeas}. The maximum reduction in the objective function achievable by moving from a feasible to an infeasible state is bounded by the maximum possible objective value, which is at most $|V|$. However, a tighter bound arises from local violations: violating a single constraint allows setting both endpoints to 0, reducing the objective by at most 2 (if both were 1) or 1 (if one was 1). More generally, for an objective with unit weights, the maximum gain from violating one constraint is 1 per variable involved. A sufficient condition is $\lambda > \max_i |w_i|$, where w_i are the objective coefficients. Here $w_i = 1$, so $\lambda > 1$ suffices. In practice, $\lambda = 2$ is chosen to provide a margin, ensuring that the energy cost of any violation dominates any potential objective improvement.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^{|V|}$ recovers the optimal vertex cover. For any feasible solution, all penalty terms vanish, so $H(x) = \sum_{i \in V} x_i$, which equals the cover size. For any infeasible solution, at least one penalty term contributes λ to the energy. Since $\lambda > 1$, the energy of any infeasible solution is strictly greater than that of any feasible solution with cover size less than λ. Specifically, if x is infeasible, there exists a feasible solution x' obtained by flipping a 0 to 1 in a violated edge; this reduces the penalty by at least λ and increases the objective by at most 1. Since $\lambda > 1$, the energy decreases, implying the global minimum cannot be infeasible. Among feasible solutions, $H(x)$ equals the objective function, so the minimum corresponds to the minimum vertex cover.

Illustrative Example. Consider a path graph with $N = 3$ vertices $V = \{0, 1, 2\}$ and edges $E = \{(0, 1), (1, 2)\}$. Here $|E| = 2$ and degrees are $\deg(0) = 1$, $\deg(1) = 2$, $\deg(2) = 1$. We set the penalty weight $\lambda = 2$.

Using the general formulas:

$$c = 2 \times 2 = 4, \tag{10}$$

$$Q_{0,0} = 1 - 2(1) = -1, \tag{11}$$

$$Q_{1,1} = 1 - 2(2) = -3, \tag{12}$$

$$Q_{2,2} = 1 - 2(1) = -1, \tag{13}$$

$$Q_{0,1} = 2, \quad Q_{1,2} = 2. \tag{14}$$

The resulting Hamiltonian is:

$$H(x) = 4 - x_0 - 3x_1 - x_2 + 2x_0x_1 + 2x_1x_2. \tag{15}$$

Evaluating $H(x)$ for candidate solutions:

- $x = (0, 1, 0)$ (cover $\{1\}$, size 1): $H = 4 - 0 - 3 - 0 + 0 + 0 = 1$.
- $x = (1, 0, 1)$ (cover $\{0, 2\}$, size 2): $H = 4 - 1 - 0 - 1 + 0 + 0 = 2$.

- $x = (0, 0, 0)$ (infeasible): $H = 4$.
- $x = (1, 1, 1)$ (cover, size 3): $H = 4 - 1 - 3 - 1 + 2 + 2 = 3$.

The global minimum is $H = 1$ at $x = (0, 1, 0)$, corresponding to the minimum vertex cover $\{1\}$.